

Paris at Ten

Emissions growth has fallen by two thirds since the Paris Agreement. We ask what the global record can, and cannot, credit the treaty for.

155 countries · 90.6% of global CO2 · 1980-2021

THE QUESTION

Something changed after Paris. Was it Paris?

The good news is real. Across the countries we can track consistently since 1980, CO2 growth fell from +2.5% a year over the Rio-to-Paris quarter century to +0.8% a year since. Worst-case scenarios have moved off the table.

But the obvious reading — that the agreement did it — is exactly the reading the data is worst at supporting. Carbon intensity has been falling for fifty years. Renewable costs collapsed on almost the same schedule as the treaty. And a slowdown in emissions can come simply from a slowdown in the economy.

So we ask a narrow question: what actually changed after 2015, and is 2015 a special year in the record — or just a year that happens to sit near one?

Why we look at intensity

Emissions have three moving parts:

economic growth + fuel mix + energy efficiency

Growth pushes emissions up. The fuel mix and efficiency are the only two things that push them down — so a policy effect has to show up in one of them before it can show up in emissions at all.

Three steps

1

Build a consistent world series

Take every country still reporting in 2021 with almost complete data back to 1980, fill the handful of gaps, and add them up. That is 155 countries and 91% of global CO₂. We stop at 2021 because most countries have not reported 2022-23 yet.

2

Let the data pick the turning point

Plot each series so a straight line means a steady percentage change per year, then fit two lines meeting at a kink. Try every possible kink year and keep the one that fits best. Nothing about a treaty enters this step.

3

Ask whether the treaty year is special

A gently curving trend makes almost any year look significant, so significance alone proves nothing. We rank 2015 against every other candidate year. A treaty earns credit only if its year beats the years around it.

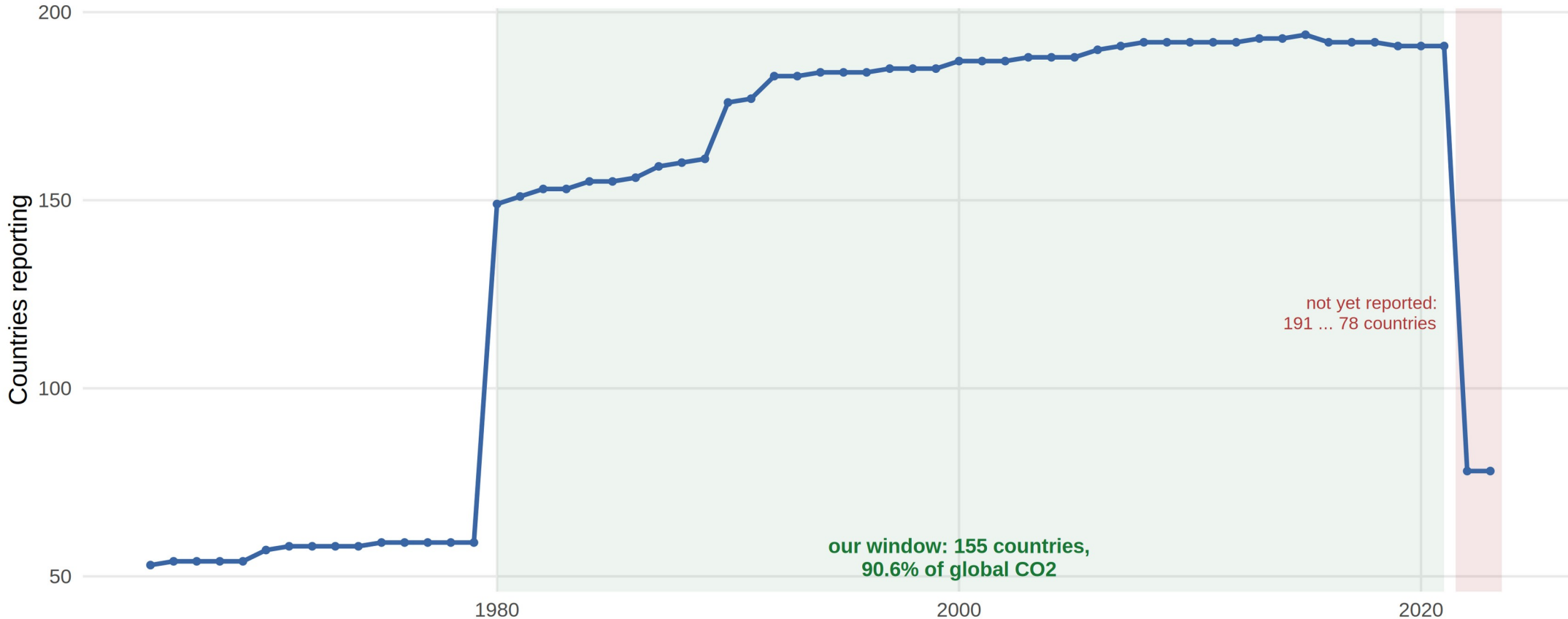
A turning point tells us when a trend changed — never, by itself, why. That is the whole method.

We use every country that reports, for as long as it reports

Takeaway: Near-complete coverage — and the last two years are dropped because most countries have not reported them.

We use every country that reports, for as long as it reports

Countries with complete emissions, energy and GDP data each year. We keep every country still reporting in 2021 with at least 90% of years present, and interpolate the 14 remaining gaps. 2022-23 are excluded: most countries have not reported them yet.

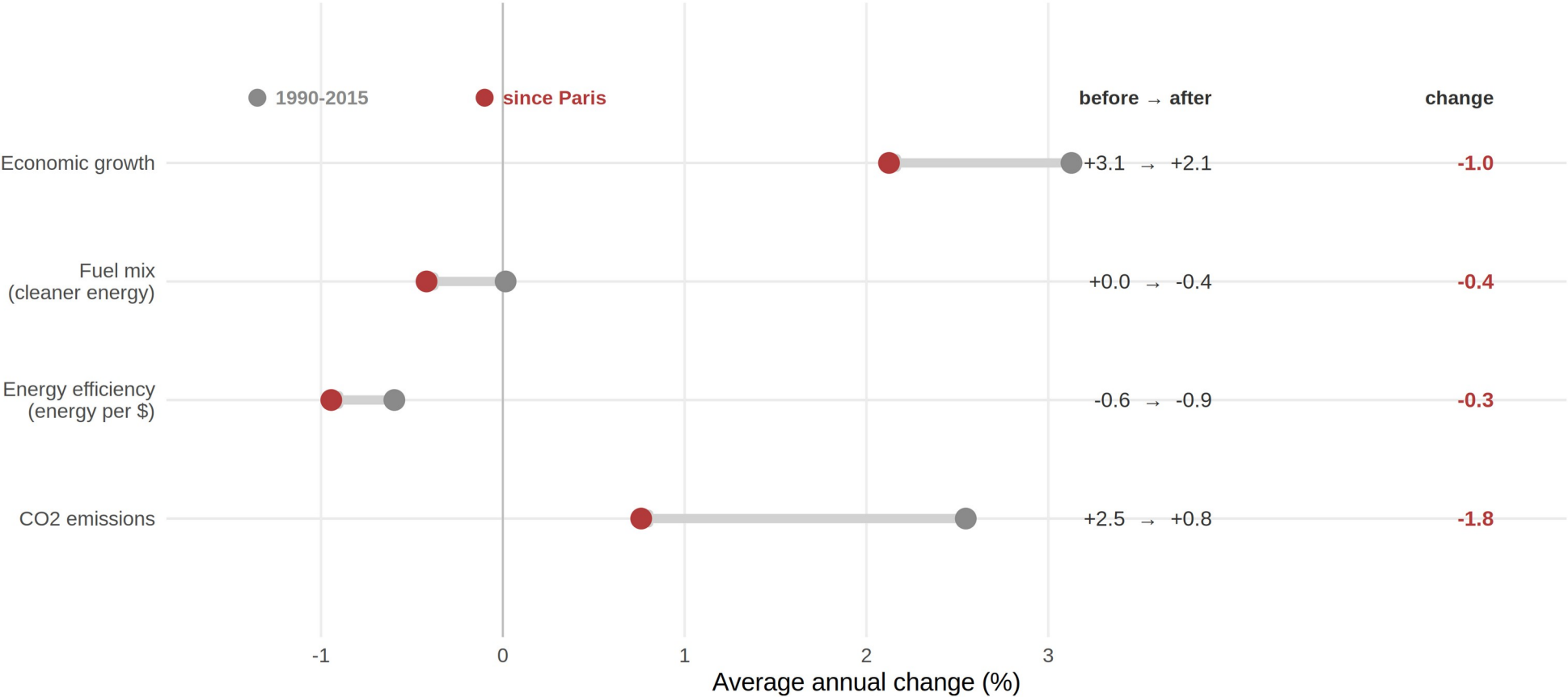


Emissions growth fell — and over half of that is slower growth

Takeaway: Of the 1.8-point slowdown, 1.0 point is a slower economy. The remaining 0.7 is real decarbonization, split between cleaner fuel and efficiency.

Emissions growth fell by two thirds - but over half of that is slower economic growth

Average annual change (%). CO2 growth = economic growth + fuel mix + energy efficiency. 155 countries, 90.6% of global CO2, 1980-2021.
Of the 1.8-point slowdown in emissions growth, 1.0 point is slower growth and 0.7 is genuine decarbonization - split between a cleaner fuel mix and efficiency.

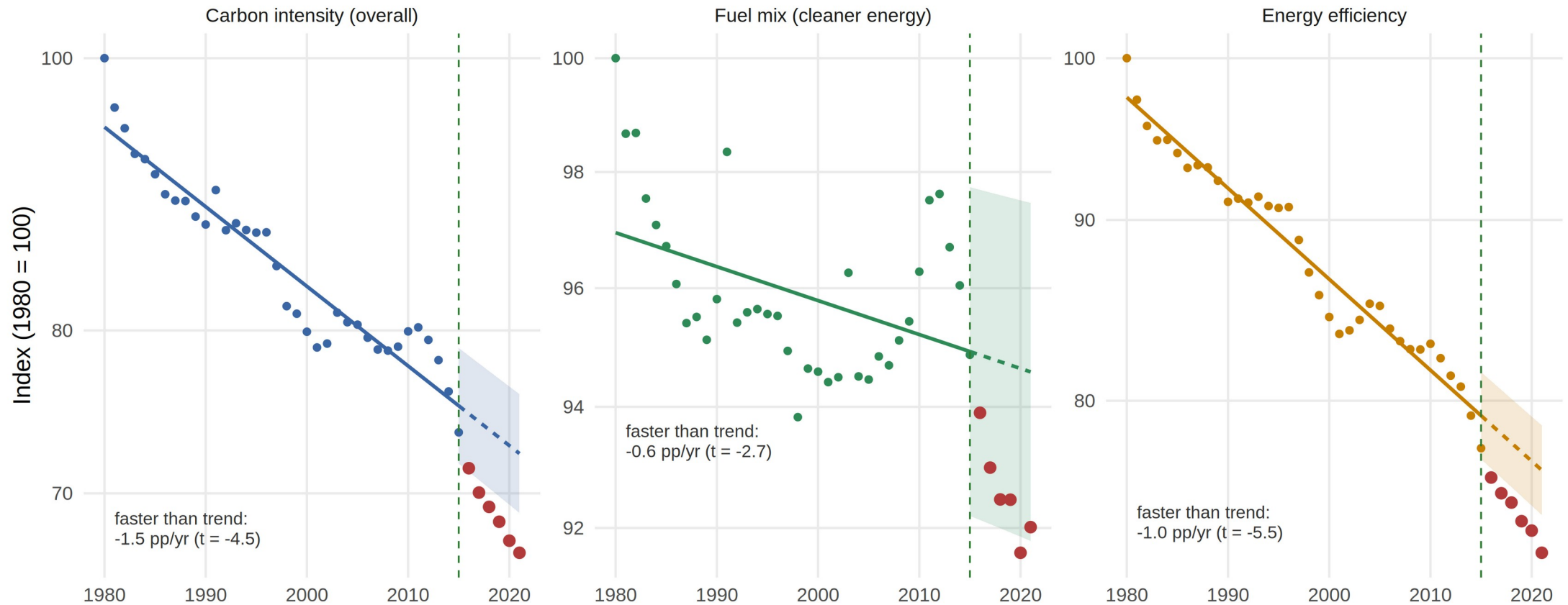


Since Paris the world has beaten its own long-run trend

Takeaway: All three measures are improving faster than 35 years of trend predicted — and the effect is largest on overall carbon intensity.

Since Paris, the world has run ahead of its own long-run trend

Each panel is indexed to 1980 = 100. The line is the 1980-2015 trend, extended past Paris with a shaded band showing where the following years should have landed if nothing had changed. Red points are what actually happened.



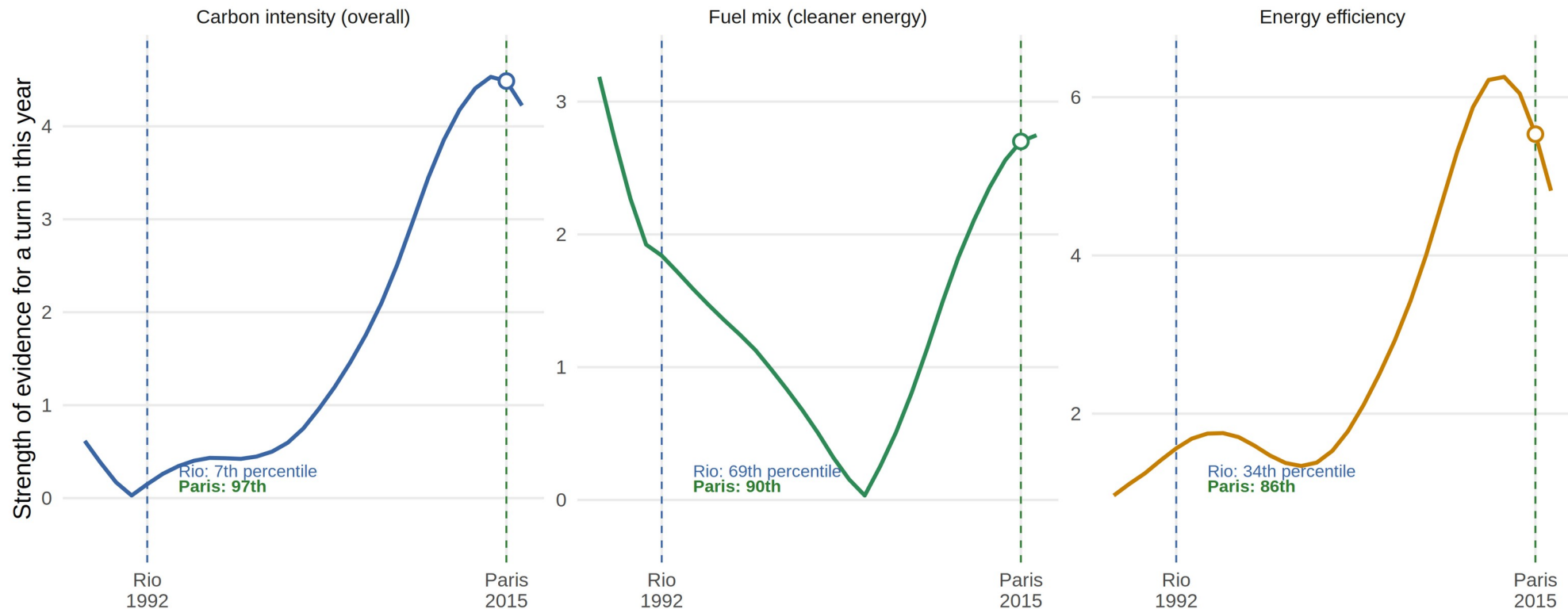
Rio fails the placebo test. Paris passes it.

Takeaway: Paris is the first treaty date the global record does not reject — it ranks in the top tenth of all candidate years.

But: the best-fitting year is 2014, not 2015, and the fuel mix prefers an older shift in the late 1980s — so the evidence is a plateau, not a spike, and it coincides with the clean-energy cost collapse.

Rio the record rejects. Paris it does not.

We put the turning point in every possible year and record how strong the evidence is. Higher means the record prefers that year. Rio ranks near the bottom of all candidate years; Paris ranks near the top. 155 countries, 1980-2021.



One result depends on which countries you include

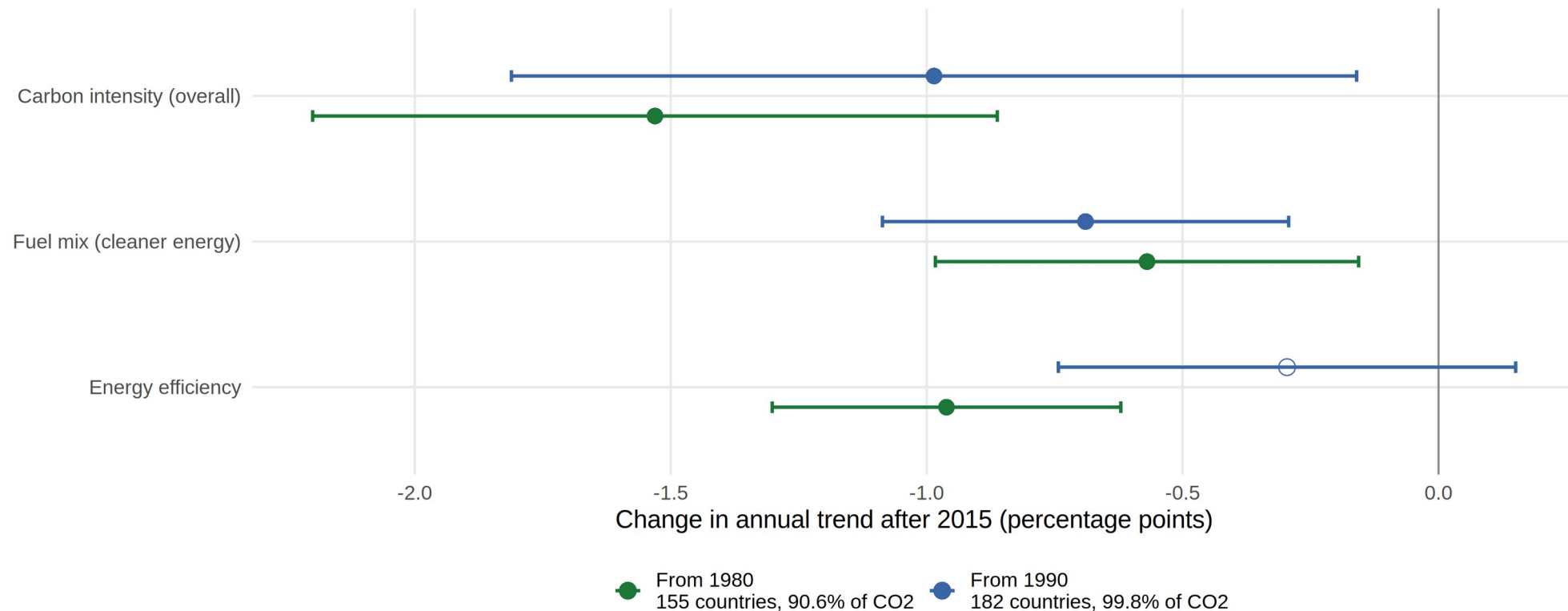
Takeaway: The overall acceleration holds on every panel. Whether efficiency contributed depends on whether post-Soviet economies are in the sample.

But: those economies cannot be carried back to 1980 — they did not exist as separate series — so this is a genuine trade-off between history and coverage, not a modelling choice.

The headline holds on either panel. The efficiency result does not.

Change in each trend after 2015, with 95% intervals. Hollow points are not statistically distinguishable from no change.

The 1990 panel adds the post-Soviet economies, whose post-1990 efficiency catch-up raises the pre-Paris baseline and absorbs the efficiency gain.



WHAT IT ADDS UP TO

Consistent with Paris. Not attributable to it.

What we can say

Emissions growth has fallen by two thirds, and roughly 0.7 points of that is genuine decarbonization rather than a slower economy. Carbon intensity is improving significantly faster than 35 years of trend predicted, on every panel we build. And 2015 ranks near the top of all candidate turning-point years — a test the 1992 Rio agreement fails badly.

What we cannot

Over half the emissions slowdown is slower economic growth, not decarbonization. The best-fitting turning point is 2014 rather than 2015, and the evidence is a plateau the test cannot resolve. That plateau coincides exactly with the collapse in clean-energy costs. And the efficiency contribution disappears once post-Soviet economies are added back.

The change is real, and it arrives exactly when cheap clean energy did. The record cannot tell those two apart.

WHAT WOULD SETTLE IT

Three things — and two of them are not at the world level

Persistence. The size of the change is already detectable — the test could pick up an effect half this large within a couple more years of data. What it cannot yet tell us is whether the new fuel-mix trend holds. If it survives to Paris+20, a one-off technology shock becomes a much weaker explanation.

The demand side. Efficiency is the more informative half precisely because it has not moved: cheap solar explains a cleaner fuel mix, but it does not explain how much energy the world uses per unit of output. If efficiency accelerates over the next decade, that is much harder to attribute to technology alone.

Disaggregation. Paris works through national pledges, so its signature should appear country by country — where series are longer, placebo tests are independent, and a documentary record exists to say what accompanied each turn. That is the companion paper.

The question we would put to the field: should treaty assessment be built on intensity turning points and placebo discipline, rather than on emissions levels and anniversaries?

LIMITS

What we are not claiming

This is not causal

A turning point identifies when a trend changed, never why. Even a perfect fit at 2015 would be a coincidence of timing. And an agreement that prevented backsliding would leave no turning point at all — so the absence of one is not the absence of an effect.

Coverage and vintage

155 countries and 91% of CO₂, with only 14 gaps interpolated. The series stops in 2021 because 2022-23 reporting is incomplete — adding those years on partial coverage would overstate the acceleration.

Which countries, and how far back

Starting in 1980 buys 35 years of pre-treaty history but drops 27 countries and 9% of global CO₂ — nearly all post-Soviet states, which have no separate 1980 series. Starting in 1990 recovers them and removes the efficiency result. Both are reported.

Production-based accounting

Emissions are counted where they are produced. At the world level that is the right frame — global production equals global consumption — but it matters for the national work that follows.

The curve is bending.

The record cannot yet tell us who bent it.

- Emissions growth fell by two thirds after Paris — but over half of that is a slower economy.
- Carbon intensity is beating 35 years of trend, significantly, on every panel we build.
- Rio fails the placebo test; Paris passes it — the first treaty date the record does not reject.
- But the best fit is 2014, not 2015, and the turn coincides with the clean-energy cost collapse.

Turning points say when · the Kaya split says through which door · placebo tests say whether the date is special